How the crisis definition was chosen (history)

Before settling on Kose–Terrones global recessions the project tried three other datings: hand-picked global and regional crisis windows on current-dollar GDP, the same windows on Maddison real GDP, and systemic banking crises from the Jordà–Schularick–Taylor database. These figures are the record of why the paper dates crises by real activity. None of their numbers is a Paper 1 result. Two captions in the July Figure Atlas PDF quoted older p-values (0.06 and 0.038) than the figures themselves show (0.038 and 0.020); the figures and the results files agree, and the values below follow them.

Status key

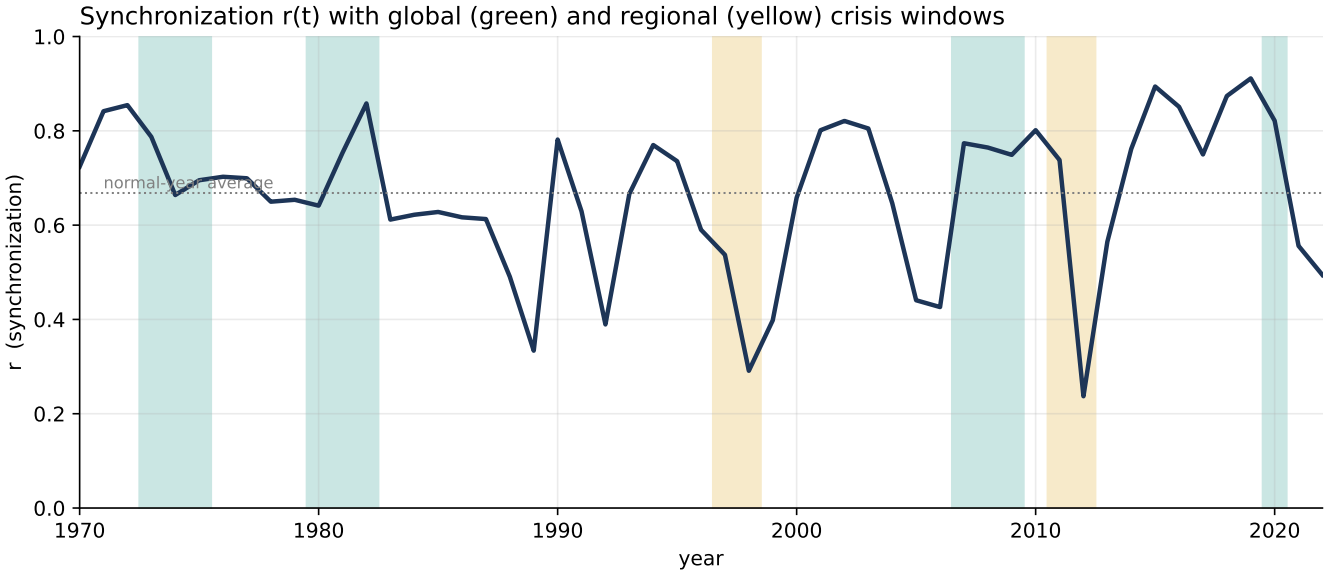
CURRENT	Paper data (Maddison 2023) and the paper's conventions; numbers match docs/methods.md.
NEEDS RELABELLING	Numbers right, but an annotation on the figure is out of date (said in the caption).
CONVENTION W	Hilbert transform applied to 1970–2022 only (convention W); the paper uses P.
SUPERSEDED	Earlier data or crisis definition; the record of how the paper got here.
DESCRIPTIVE	Atlas material from the Global Macro Database; describes, does not test.
UNVERIFIED SERIES	The series drawn matches no stored results file; redraw before use.

In this book

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F3.2	Current dollars: each global crisis, zoomed	SUPERSEDED
F3.3	Current dollars: composite around global-crisis onsets	SUPERSEDED
F3.4	Current dollars: global-crisis windows against randomly placed windows	SUPERSEDED
F3.5	Current dollars: mean synchronization by period type	SUPERSEDED
F3.6	Real GDP: synchronization with the same hand-picked windows	SUPERSEDED
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F3.14	Banking crises: the one crisis that went global began at peak synchronization	SUPERSEDED
F3.15	Two crisis definitions, two verdicts	SUPERSEDED
F3.16	Synchronization–crisis correlation by timing, both definitions	SUPERSEDED

F3.1 SUPERSEDED

Current dollars: synchronization with hand-picked crisis windows

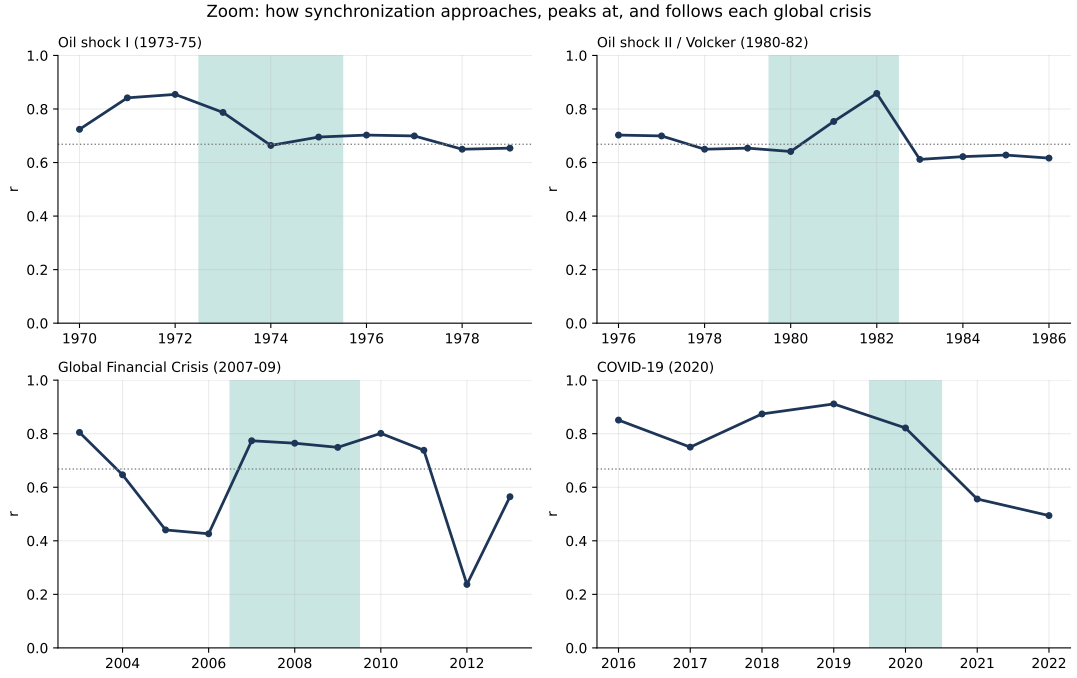


$R(t)$ on current-dollar GDP with four global crisis windows (green: 1973–75, 1980–82, 2007–09, 2020) and two regional ones (yellow: 1997–98, 2011–12).

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/crisis_analysis.py → crisis_results.json (reproduces exactly) · **Data** World Bank current US$
File figures/03_crisis_dating_history/F3-01_cfig1_overview.pdf

F3.2 SUPERSEDED

Current dollars: each global crisis, zoomed

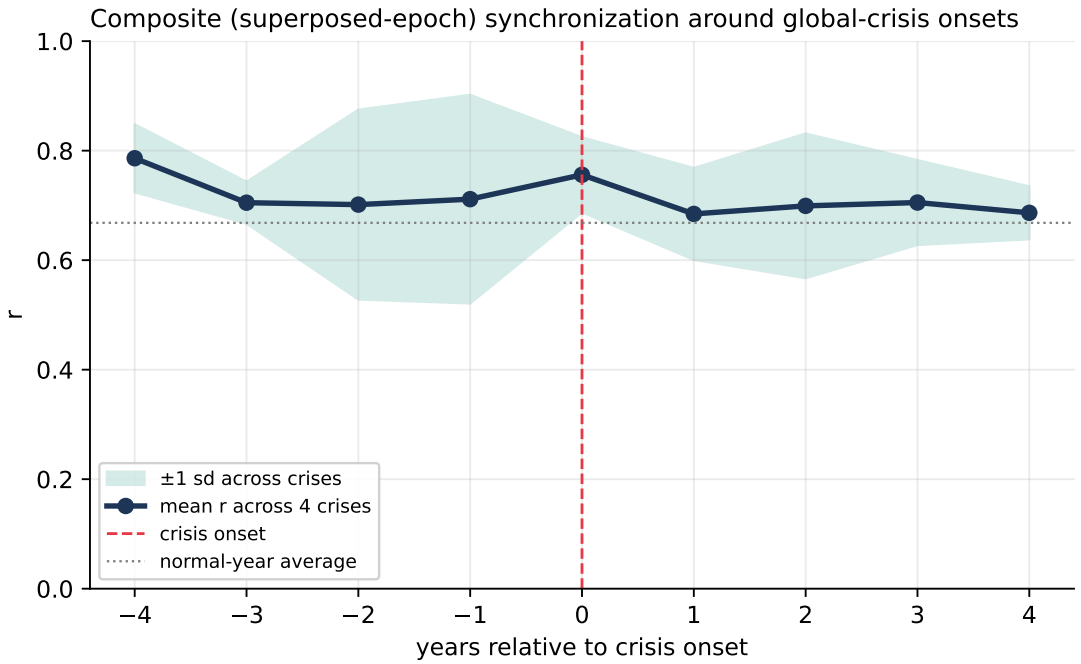


Synchronization before, during and after each of the four global windows.

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/crisis_analysis.py → crisis_results.json (reproduces exactly) · **Data** World Bank current US$
File figures/03_crisis_dating_history/F3-02_cfig2_zoom.pdf

F3.3 SUPERSEDED

Current dollars: composite around global-crisis onsets

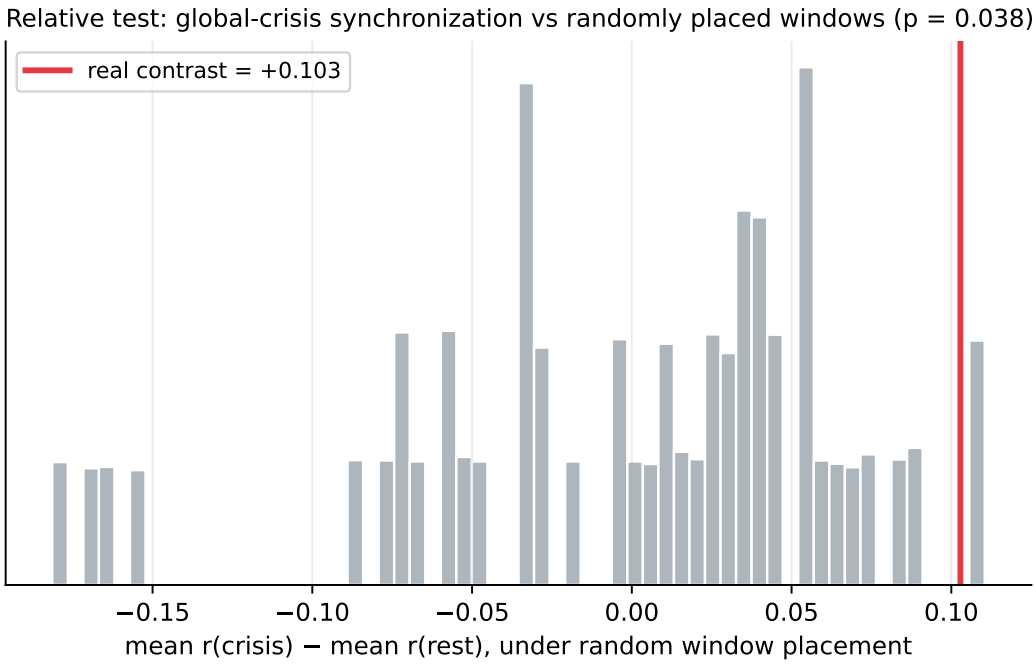


Mean R across the four global crises, ±1 s.d., against the normal-year average.

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/crisis_analysis.py → crisis_results.json (reproduces exactly) · **Data** World Bank current US$
File figures/03_crisis_dating_history/F3-03_cfig3_composite.pdf

F3.4 SUPERSEDED

Current dollars: global-crisis windows against randomly placed windows

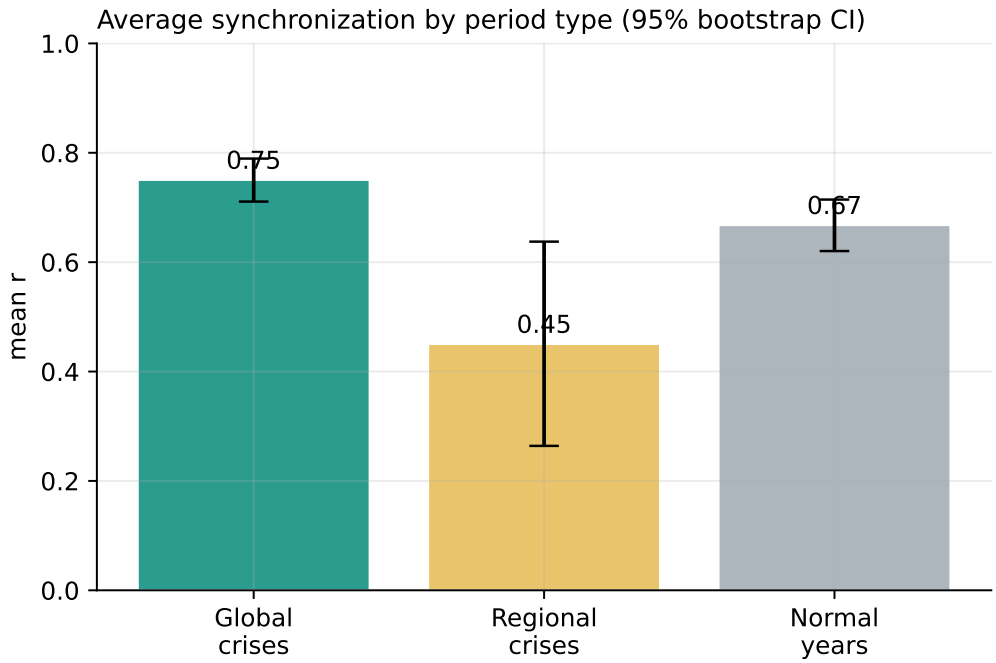


The contrast between mean R in global-crisis years and all other years, against 20,000 random placements of windows of the same lengths. (The July Figure Atlas caption said p ≈ 0.06; the figure and crisis_results.json say 0.038.)

Numbers Real contrast +0.103; p = 0.038.
Made by paper1/scripts/eraA_claude_ai/crisis_analysis.py → crisis_results.json (reproduces exactly) · **Data** World Bank current US$
File figures/03_crisis_dating_history/F3-04_cfig4_permutation.pdf

F3.5 SUPERSEDED

Current dollars: mean synchronization by period type

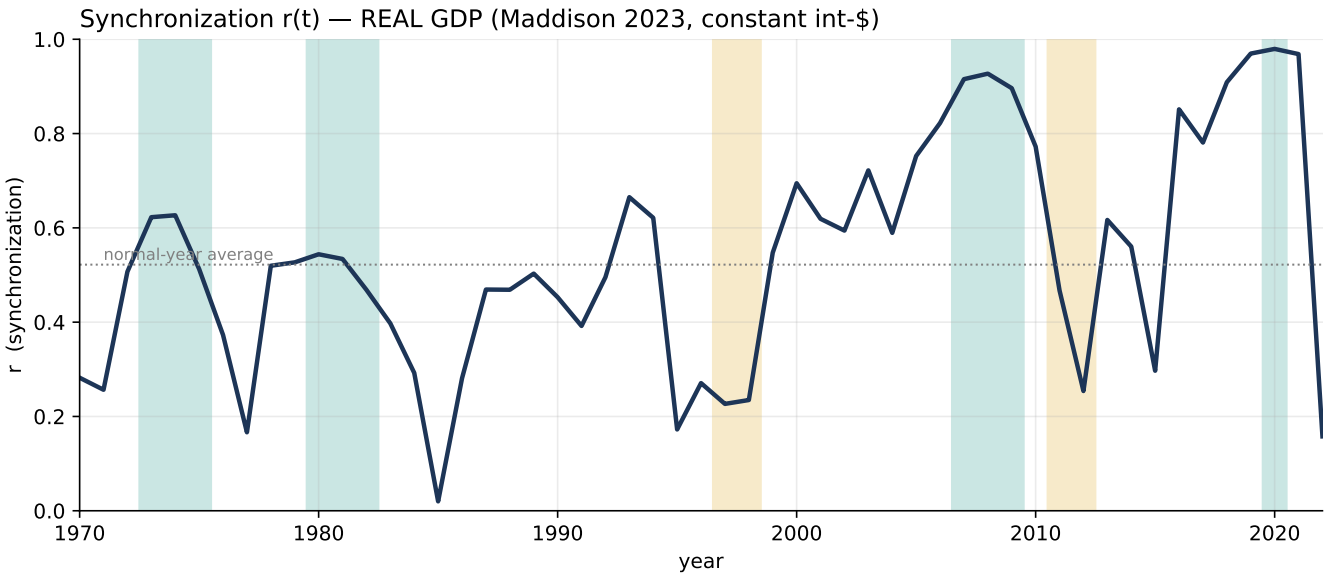


Mean R with 95% bootstrap intervals.

Numbers Global 0.75; regional 0.45; normal 0.67.
Made by paper1/scripts/eraA_claude_ai/crisis_analysis.py → crisis_results.json (reproduces exactly) · **Data** World Bank current US$
File figures/03_crisis_dating_history/F3-05_cfig5_groups.pdf

F3.6 SUPERSEDED

Real GDP: synchronization with the same hand-picked windows

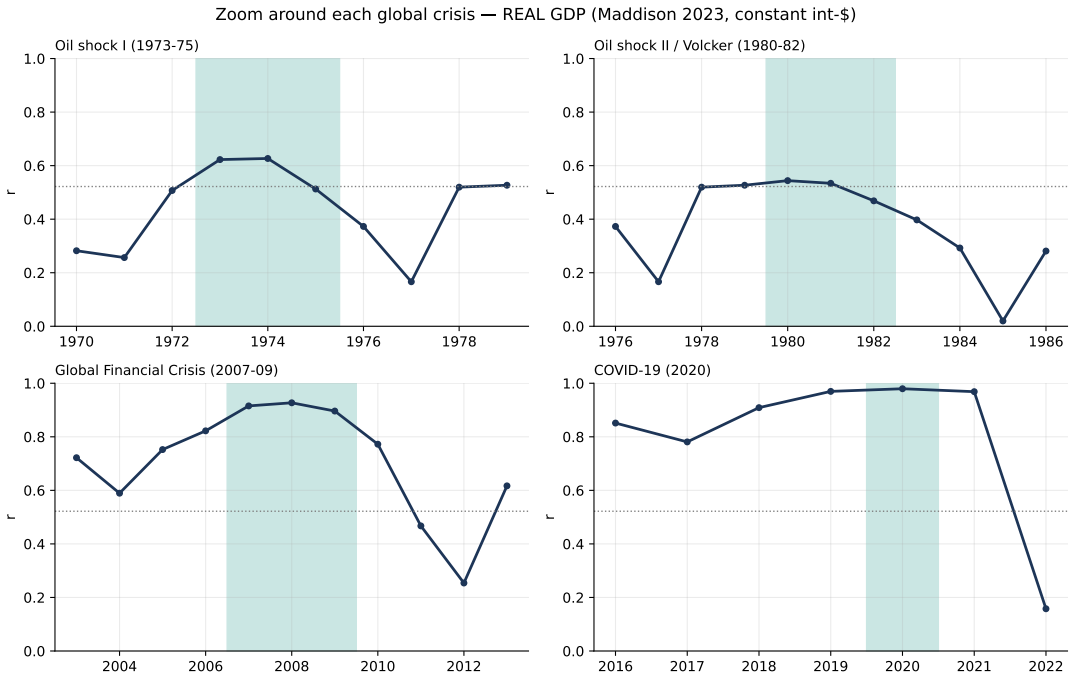


The same windows on Maddison real GDP per capita (convention P).

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/real_gdp_analysis.py → crisis_results_real.json (reproduces exactly) · **Data** Maddison 2023
File figures/03_crisis_dating_history/F3-06_rfig1_overview.pdf

F3.7 SUPERSEDED

Real GDP: each global crisis, zoomed

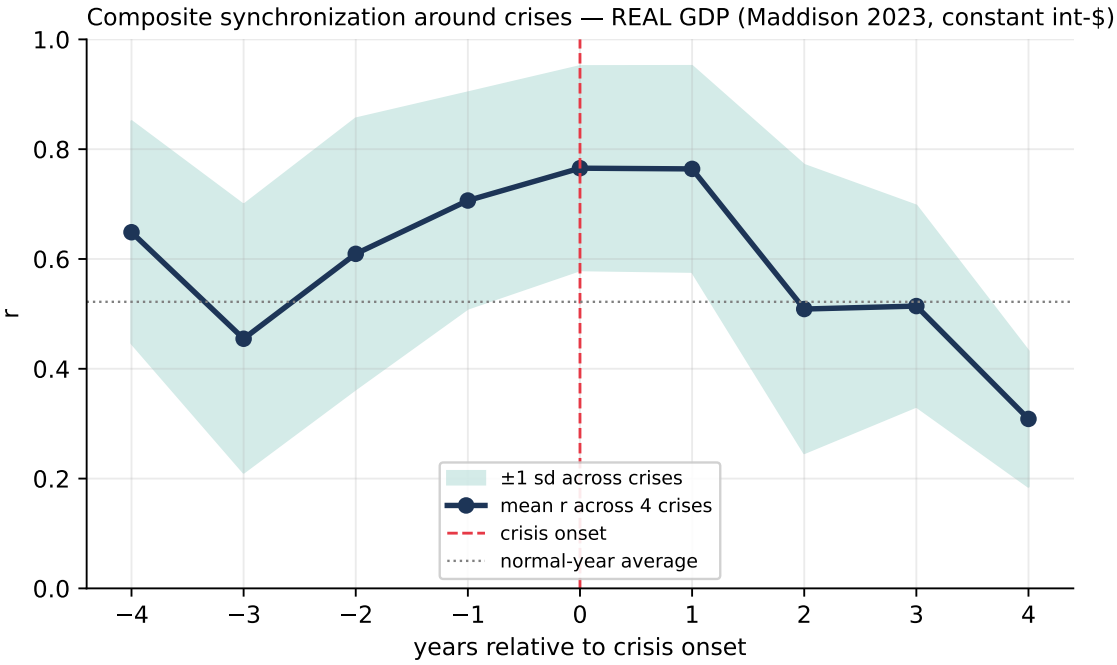


The 2007–09 and 2020 shapes are cleaner in real terms; the oil shocks are muted.

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/real_gdp_analysis.py → crisis_results_real.json (reproduces exactly) · **Data** Maddison 2023
File figures/03_crisis_dating_history/F3-07_rfig2_zoom.pdf

F3.8 SUPERSEDED

Real GDP: composite around global-crisis onsets

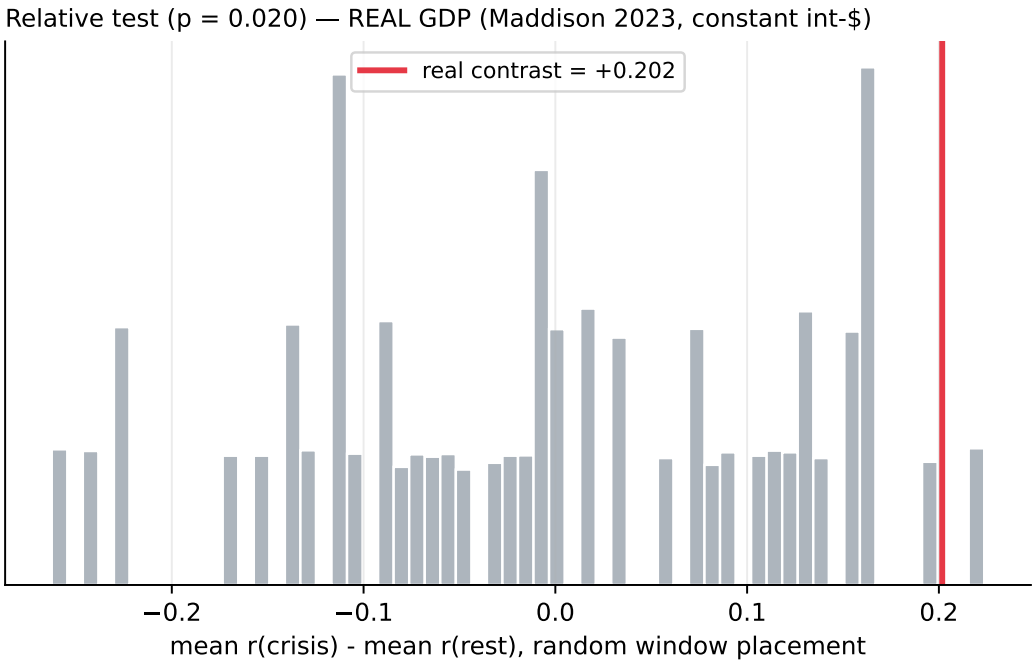


Mean r across the four global crises, ±1 s.d.

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/real_gdp_analysis.py → crisis_results_real.json (reproduces exactly) · **Data** Maddison 2023
File figures/03_crisis_dating_history/F3-08_rfig3_composite.pdf

F3.9 SUPERSEDED

Real GDP: global-crisis windows against randomly placed windows

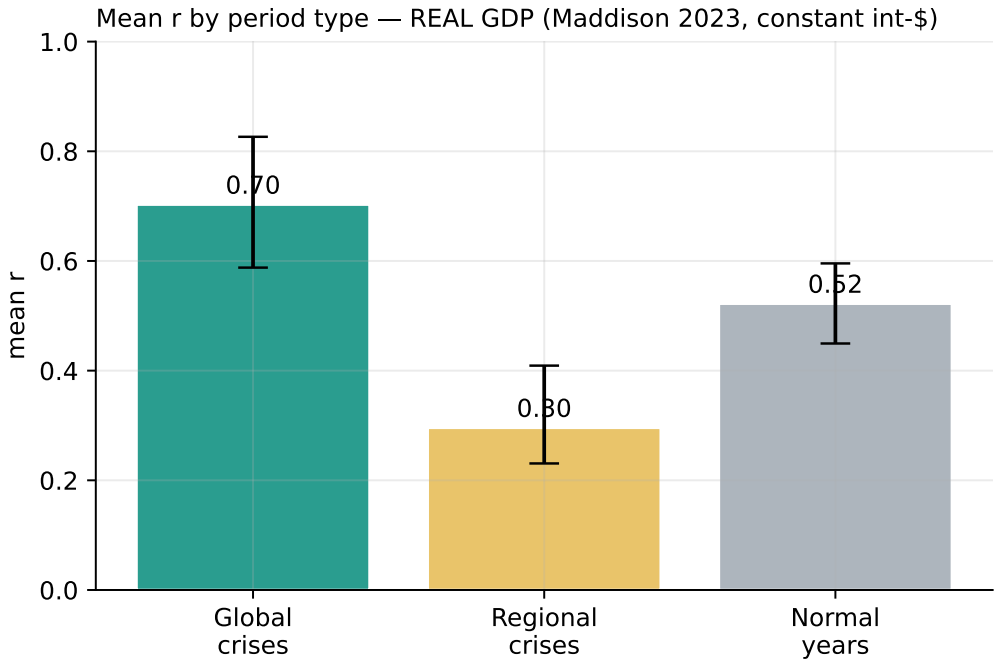


As F3.4 on real GDP. (The July Figure Atlas caption said p = 0.038; the figure and crisis_results_real.json say 0.020.)

Numbers Real contrast +0.202; p = 0.020.
Made by paper1/scripts/eraA_claude_ai/real_gdp_analysis.py → crisis_results_real.json (reproduces exactly) · **Data** Maddison 2023
File figures/03_crisis_dating_history/F3-09_rfig4_permutation.pdf

F3.10 SUPERSEDED

Real GDP: mean synchronization by period type

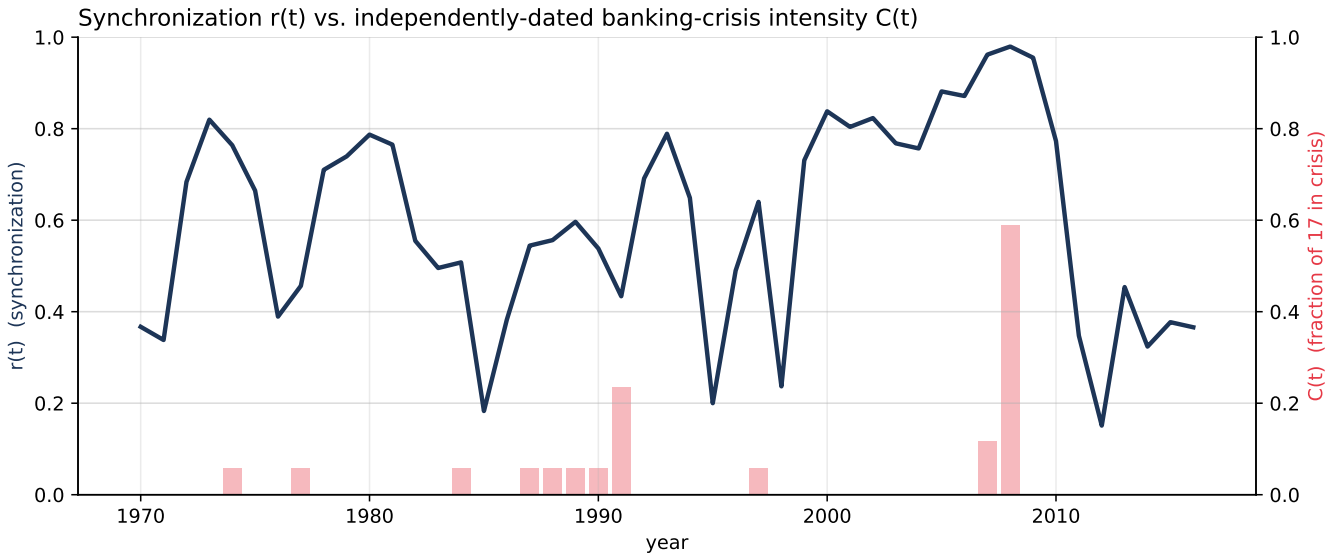


Mean r with 95% bootstrap intervals.

Numbers Global 0.70; regional 0.30; normal 0.52.
Made by paper1/scripts/eraA_claude_ai/real_gdp_analysis.py → crisis_results_real.json (reproduces exactly) · **Data** Maddison 2023
File figures/03_crisis_dating_history/F3-10_rfig5_groups.pdf

F3.11 SUPERSEDED

Banking crises: synchronization against banking-crisis intensity $C(t)$

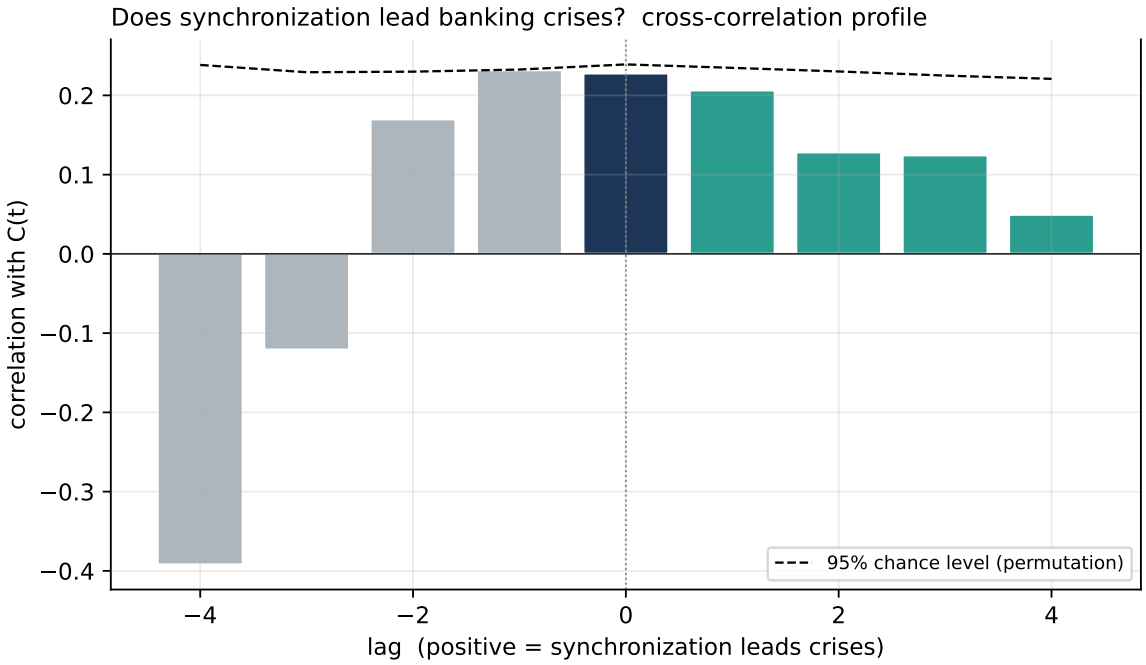


$r(t)$ for the 17 JST economies (1970–2016) against $C(t)$, the share of them in a systemic banking crisis. Dating crises from bank failures breaks the circularity of hand-picked windows, but the crises cluster on one global event (2008).

Numbers Descriptive.
Made by paper1/scripts/eraA_claude_ai/crisis_dating_test.py → crisis_dating_results.json (not re-verified: needs jst.xlsx) · **Data** Jordà–Schularick–Taylor Macrohistory Database
File figures/03_crisis_dating_history/F3-11_dfig1_r_vs_C.pdf

F3.12 SUPERSEDED

Banking crises: lead–lag profile

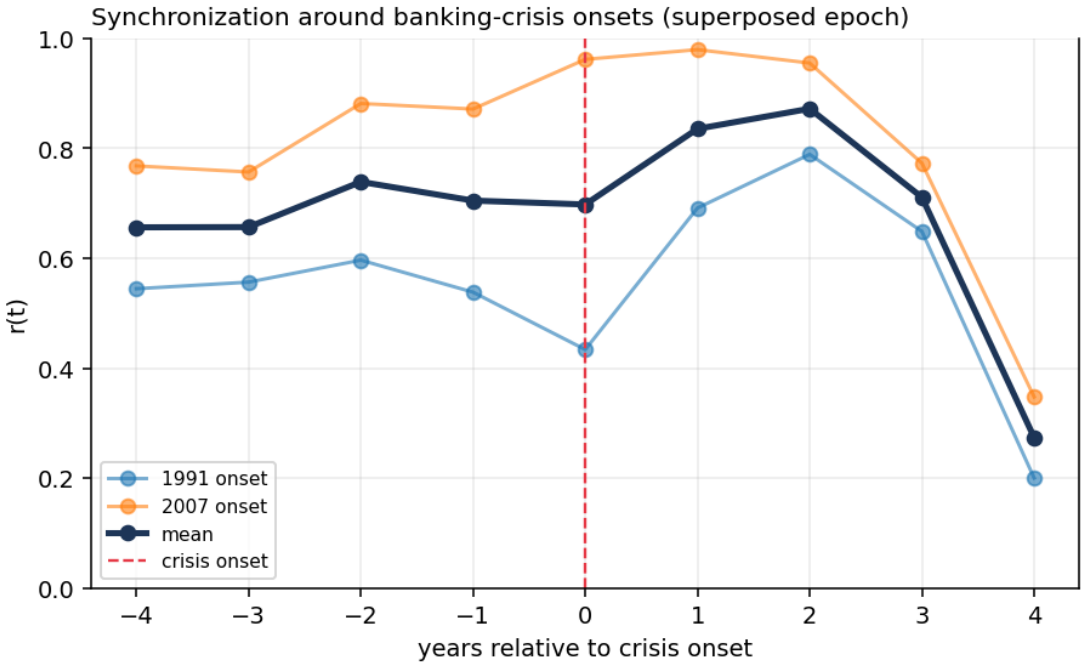


Cross-correlation of $r(t)$ with $C(t)$; positive lags mean synchronization comes first. No lag clears the permutation chance level, because there is essentially one global banking crisis.

Numbers Same year +0.23 (p = 0.106); lead-1 +0.21 (0.149); lead-2 +0.13 (0.298).
Made by crisis_dating_test.py · **Data** JST
File figures/03_crisis_dating_history/F3-12_dfig2_leadlag.pdf

F3.13 SUPERSEDED

Banking crises: synchronization around banking-crisis onsets

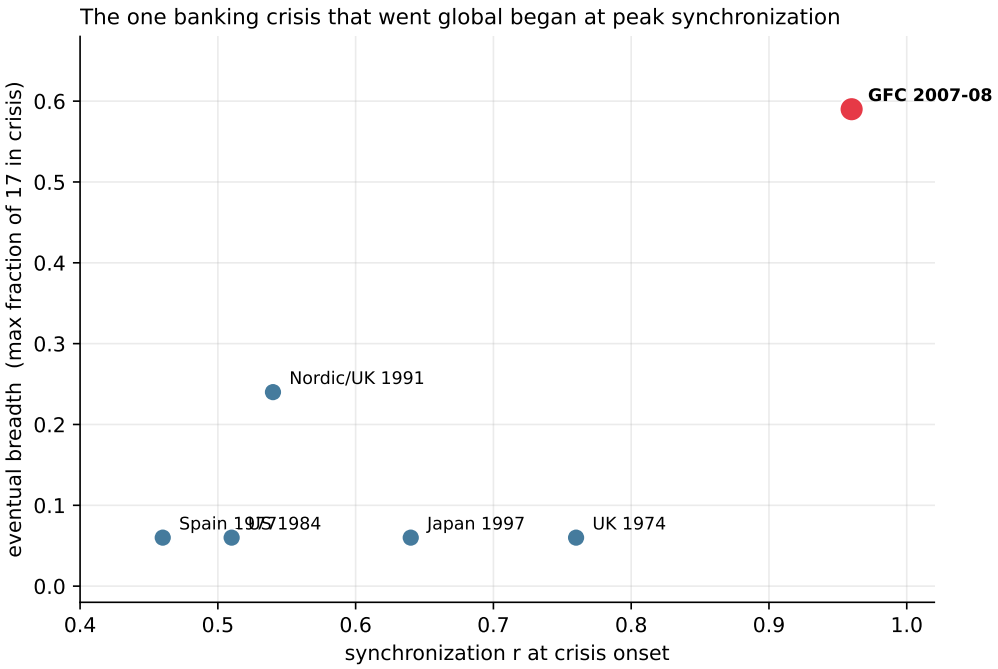


Superposed epoch around the 1991 and 2007 banking-crisis onsets and their mean. Recovered from the July Figure Atlas PDF; the original file was not kept.

Numbers Descriptive.
Made by crisis_dating_test.py · **Data** JST
File figures/03_crisis_dating_history/F3-13_dfig3_epoch.png

F3.14 SUPERSEDED

Banking crises: the one crisis that went global began at peak synchronization

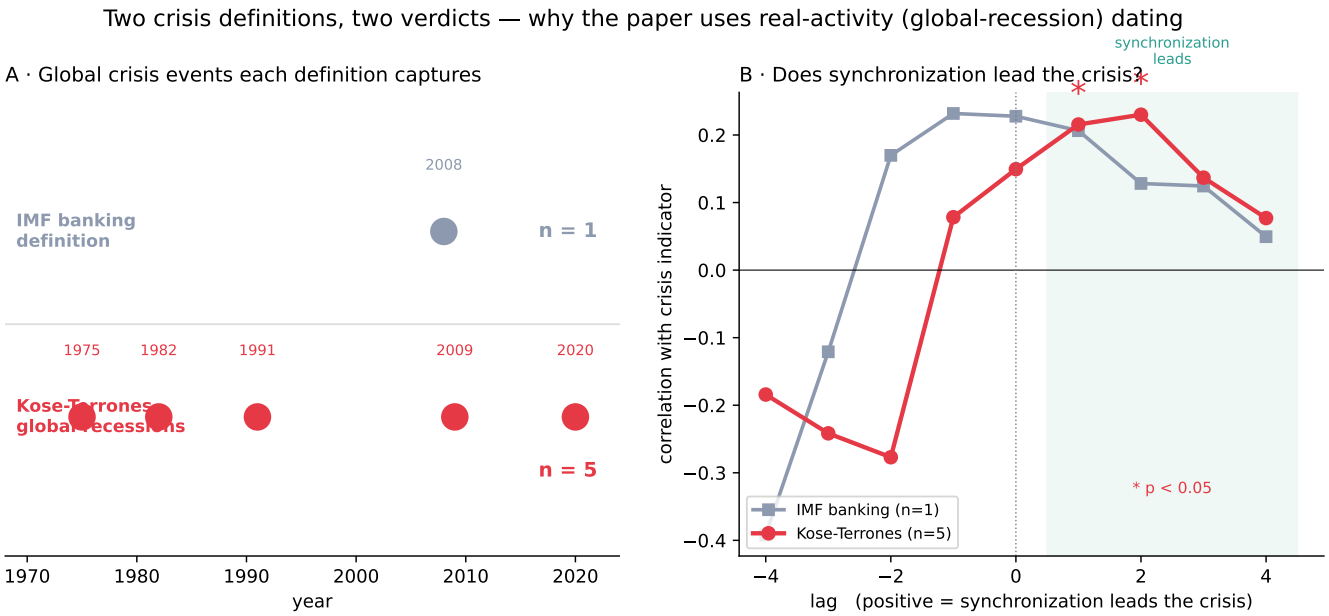


Synchronization at each banking-crisis onset against how widely the crisis eventually spread. The 2007–08 crisis began near peak synchronization and reached most of the 17; crises that began at ordinary synchronization stayed contained. One event, so this is an anecdote, not a test.

Numbers 2007–08: onset r ≈ 0.96, breadth ≈ 0.59.
Made by crisis_dating_test.py · **Data** JST
File figures/03_crisis_dating_history/F3-14_dfig4_onset_breadth.pdf

F3.15 SUPERSEDED

Two crisis definitions, two verdicts

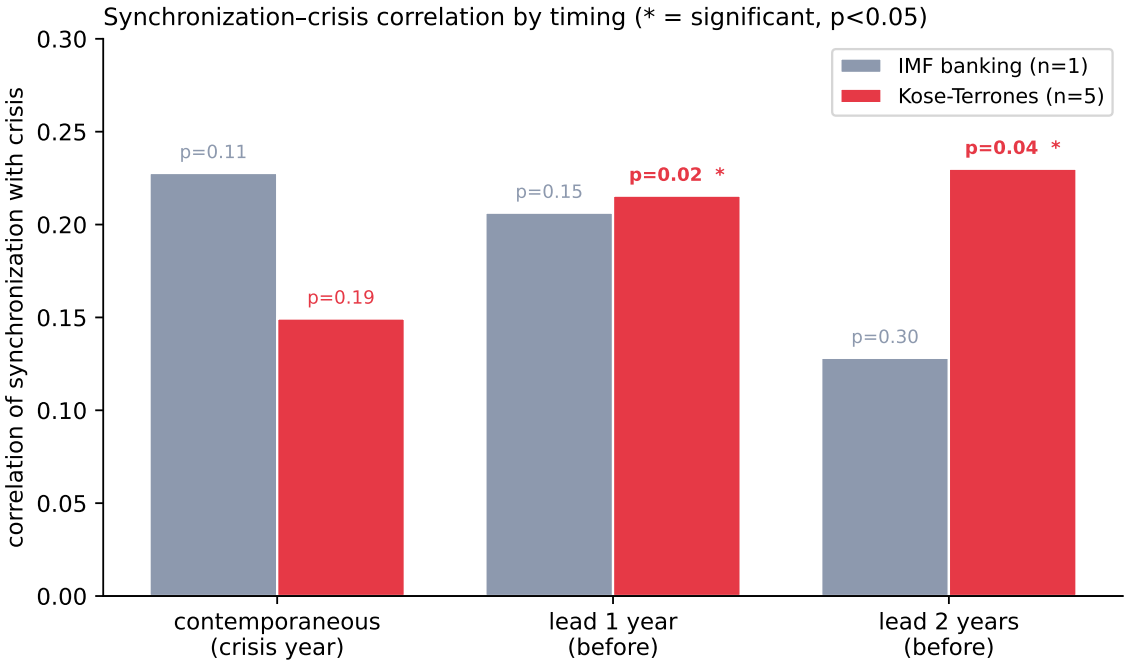


(A) The banking definition captures one global event (2008); Kose–Terrones captures five. (B) Their lead–lag profiles: only the real-activity definition shows a significant lead. This is the figure that justifies the paper's choice of crisis dating.

Numbers Kose–Terrones significant at lead-1 and lead-2; banking never.
Made by comparison of the two results files (plotting script not kept) · **Data** Maddison 2023 and JST
File figures/03_crisis_dating_history/F3-15_compare_definitions.pdf

F3.16 SUPERSEDED

Synchronization–crisis correlation by timing, both definitions



Correlations at the crisis year and one and two years before, for the banking definition (n = 1) and Kose–Terrones (n = 5).

Numbers Banking: p = 0.11, 0.15, 0.30. Kose–Terrones: p = 0.19, 0.02, 0.04.
Made by as F3.15 · **Data** Maddison 2023 and JST
File figures/03_crisis_dating_history/F3-16_compare_leadstats.pdf